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QUESTION NO 1

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp12
{
    internal class Product
    {
        public int pid { get; set; }

        public String pname { get; set; }

        public String category { get; set; }

        public String compname { get; set; }

        public int price { get; set; }

        public int stock_q { get; set; }

    }
}

using System;
using System.Collections.Generic;
using System.Data;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp12
{
    internal class Program
    {
        static List<Product> products = new List<Product>();
        static void Main(string[] args)

        {
            int choice;

            do
            {
```

```

Console.WriteLine("\tTuck Shop Inventory Management \t");
Console.WriteLine("1.Add Product");
Console.WriteLine("2.Update Product");
Console.WriteLine("3.Remove Product");
Console.WriteLine("4.Search Product by ID");
Console.WriteLine("5.Search Products by Price Range");
Console.WriteLine("6.Display All Products");
Console.WriteLine("7.Exit");
choice = int.Parse(Console.ReadLine());
switch (choice)
{
    case 1:
        AddProduct();
        break;
    case 2:
        UpdateProduct();
        break;
    case 3:
        RemoveProduct();
        break;
    case 4:
        SearchByID();
        break;
    case 5:
        SearchByPriceRange();
        break;
    case 6:
        DisplayAll();
        break;
    case 7:
        Console.WriteLine("Exiting Program!!");
        break;
    default:
        Console.WriteLine("Invalid choice");
        break;
}

} while (choice != 7);
}

static void AddProduct()
{
    Product p = new Product();
    Console.WriteLine("Enter Product Id");
    p.pid = int.Parse(Console.ReadLine());
    Console.WriteLine("Enter Product Name");
    p.pname = Console.ReadLine();
    Console.WriteLine("Enter Product category");
    p.category = Console.ReadLine();
    Console.WriteLine("Enter Product price");
    p.price = int.Parse(Console.ReadLine());
    Console.WriteLine("Enter Product stock quantity");
    p.stock_q = int.Parse(Console.ReadLine());
    Console.WriteLine("Enter Product company name");
    p.compname = Console.ReadLine();
    products.Add(p);
}

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    }
    static void UpdateProduct()
    {
        Console.WriteLine("Enter Product Id");
        int id = int.Parse(Console.ReadLine());

        Product found = products.Find(p => p.pid == id);
        if (found != null)
        {
            Console.WriteLine("Enter Product Name");
            found.pname = Console.ReadLine();
            Console.WriteLine("Enter Product category");
            found.category = Console.ReadLine();
            Console.WriteLine("Enter Product price");
            found.price = int.Parse(Console.ReadLine());
            Console.WriteLine("Enter Product stock quantity");
            found.stock_q = int.Parse(Console.ReadLine());
            Console.WriteLine("Enter Product company name");
            found.compname = Console.ReadLine();
        }
        else
        {
            Console.WriteLine("Product not found");
        }
    }

    static void RemoveProduct()
    {
        Console.WriteLine("Enter Product Id");
        int id = int.Parse(Console.ReadLine());

        Product found = products.Find(p => p.pid == id);
        if (found != null)
        {
            products.Remove(found);
            Console.WriteLine("product is sucessfully removed");
        }
        else
        {
            Console.WriteLine("Product not found");
        }
    }

    static void SearchByID()
    {
        Console.WriteLine("Enter Product Id");
        int id = int.Parse(Console.ReadLine());

        Product found = products.Find(p => p.pid == id);
        if (found != null)
        {
            Console.WriteLine(" Product Id" + found.pid);
            Console.WriteLine(" Product Name" + found.pname);
            Console.WriteLine(" Product category" + found.category);
            Console.WriteLine(" Product price" + found.price);
            Console.WriteLine("Product stock quantity" + found.stock_q);
            Console.WriteLine(" Product company name" + found.compname);
        }
    }

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    }

    else
    {
        Console.WriteLine("Product not found");
    }
}

static void SearchByPriceRange()
{
    Console.Write("Enter Minimum Price: ");
    double min = double.Parse(Console.ReadLine());
    Console.Write("Enter Maximum Price: ");
    double max = double.Parse(Console.ReadLine());

    var products = Product.FindAll(x => x.Price >= min && x.Price <= max);

    if (products.Count > 0)
    {
        Console.WriteLine("Products in this range:");
        foreach (var p in products)
        {
            Console.WriteLine($"{p.ProductName} - Rs. {p.Price}");
        }
    }
    else
    {
        Console.WriteLine(" No products found in this range.");
    }
}

static void DispalyAll()
{
    if (products.Count == 0)
    {
        Console.WriteLine("no product avalible");
    }
    else
    {
        foreach (Product p in products)
        {
            Console.WriteLine(" Product Id" + p.pid);
            Console.WriteLine(" Product Name" + p.pname);
            Console.WriteLine(" Product category" + p.category);
            Console.WriteLine(" Product price" + p.price);
            Console.WriteLine("Product stock quantity" + p.stock_q);
            Console.WriteLine(" Product company name" + p.compname);
        }
    }
}
}
}
}
}

```

QUESTION NO 2:

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace WindowsFormsApp7
{
    internal class Student
    {

        public int ID { get; set; }
        public string Name { get; set; }
        public string City { get; set; }
        public DateTime DOB { get; set; }
        public string Gender { get; set; }
        public string Department { get; set; }
        public string Course { get; set; }

    }
}

```

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.Xml.Linq;

namespace WindowsFormsApp7
{
    public partial class Form1 : Form
    {
        List<Student> students = new List<Student>()
        Dictionary<string, List<string>> departmentCourses = new Dictionary<string, List<string>>();

        public Form1()
        {
            InitializeComponent();

            string[] cities = new string[]
            {
                "Lahore", "Karachi", "Islamabad", "Rawalpindi",
                "Multan", "Faisalabad", "Peshawar", "Quetta"
            };
            cmbcity.DataSource = cities;
            List<string> departments = new List<string> { "CS", "Math", "Others" };
            cmbdep.DataSource = departments;

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departmentCourses.Add("CS", new List<string> { "C#", "C++", "Java", "Python",
"Database" });
departmentCourses.Add("Math", new List<string> { "Algebra", "Calculus",
"Geometry", "Statistics", "Trigonometry" });
departmentCourses.Add("Others", new List<string> { "English", "Urdu",
"Islamiyat", "Pak Studies", "History" });

cmbdep.SelectedIndexChanged += cmbDept_SelectedIndexChanged;
}

private void cmbDept_SelectedIndexChanged(object sender, EventArgs e)
{
cmbcourse.DataSource = null;
string selectedDept = cmbdep.Text;

if (departmentCourses.ContainsKey(selectedDept))
{
cmbcourse.DataSource = departmentCourses[selectedDept];
}
}

private void btnInsert_Click(object sender, EventArgs e)
{
if (!int.TryParse(txtID.Text, out int id))
{
MessageBox.Show("ID must be numeric!");
return;
}

if (string.IsNullOrEmpty(txtName.Text))
{
MessageBox.Show("Name cannot be empty!");
return;
}

if (cmbCity.SelectedIndex == -1)
{
MessageBox.Show("Please select a city!");
return;
}

if (dobPicker.Value > DateTime.Now)
{
MessageBox.Show("Date of birth cannot be in future!");
return;
}
if (gender == "")
{
MessageBox.Show("Please select gender!");
return;
}

Student s = new Student()
{
ID = id,
Name = txtName.Text,

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City = cmbCity.Text,
DOB = dobPicker.Value,
Gender = gender,
Department = cmbDept.Text,
Course = cmbCourse.Text
};

students.Add(s);
MessageBox.Show("Student added successfully!");
}
private void btnUpdate_Click(object sender, EventArgs e)
{
if (!int.TryParse(txtID.Text, out int id))
{
MessageBox.Show(" Enter valid numeric ID!");
return;
}

Student s = students.Find(x => x.ID == id);
if (s == null)
{
MessageBox.Show(" Student not found");
return;
}

s.Name = txtName.Text;
s.City = cmbCity.Text;
s.DOB = dobPicker.Value;
s.Gender = radioMale.Checked ? "Male" : "Female";
s.Department = cmbDept.Text;
s.Course = cmbCourse.Text;

MessageBox.Show("Student updated ");
}
private void btnDelete_Click(object sender, EventArgs e)
{
if (!int.TryParse(txtID.Text, out int id))
{
MessageBox.Show("Enter valid numeric ID");
return;
}

Student s = students.Find(x => x.ID == id);
if (s == null)
{
MessageBox.Show("Student not found");
return;
}

students.Remove(s);
MessageBox.Show("Student deleted ");
}
private void btnGet_Click(object sender, EventArgs e)
{
if (!int.TryParse(txtID.Text, out int id))
{
MessageBox.Show("Enter valid numeric ID");
return;
}

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}

Student s = students.Find(x => x.ID == id);
if (s == null)
{
    MessageBox.Show("Student not found");
    return;
}

txtName.Text = s.Name;
cmbCity.Text = s.City;
dobPicker.Value = s.DOB;
cmbDept.Text = s.Department;
cmbCourse.Text = s.Course;

if (s.Gender == "Male") radioMale.Checked = true;
else radioFemale.Checked = true;
}
}

}
```